ECA5604 – 5G/6G Communication Systems

CO1 - INTEGRATED ENGINEERING ANALYSIS AND PLANNING ASSIGNMENT

Evolution, Cellular Planning, Spectrum Use and Interference Management for a Regional Mobile Network

Item	Details
Course Outcome	CO1 (Bloom's K4 – Analyze, K5 – Evaluate, K6 – Create)
Maximum Marks	100
Student Name / Reg. No.	Manoj K / 192312407
Department / Section	ECE
Submission Date	28-10-2026

Contents: Key points · Given data and assumptions · Stage 1 Evolution of mobile systems · Stage 2 Cell coverage and frequency reuse · Stage 3 Traffic capacity · Stage 4 Mobility and handoff · Stage 5 Interference and spectrum bands · Stage 6 Spectrum coordination and recommendation · Calculation verification (script) · Final verification table · Final system design · Final engineering decision · References

Key Points

- **Evolution:** 1G AMPS (analog FDMA, 30 kHz) → 2G GSM (digital, 200 kHz × 8 TDMA slots) → 3G UMTS (5 MHz CDMA) → 4G LTE (OFDMA + MIMO) → 5G NR (eMBB/URLLC/mMTC, mid-band + mmWave, beamforming) → 6G vision (THz, AI-native, ISAC). Each step made spectrum more efficiently used and shifted interference management from fixed planning to dynamic, intelligent control.
- **1G numbers:** λ(800 MHz) = 0.375 m; idealized channels = 10 MHz / 30 kHz = 333 (guard bands, control channels and duplexing ignored; practical count is lower).
- **Geometry:** one hexagonal cell = 2.598 r^2 = 10.39 km²; 19-cell cluster = 197.45 km²; for N = 7 the reuse fraction is 1/7 = 0.143 and $D = r\sqrt{(3N)}$ = 9.17 km (D/r = 4.58).
- **Traffic:** A = 1000 × 0.08 = 80 Erlang. Erlang B with C = 100 gives 0.40 % blocking; the smallest C for ≤ 2 % is **92** (1.92 %). One 100-channel group is sufficient (first-order); it saturates at about 88 E.
- **Mobility:** 60 km/h = 16.7 m/s, so the 6 s overlap is 100 m; with a 1 km route cell length there is **1 hard handoff per minute**; single-handoff drop risk ≈ 1 % (≈ 4.9 % over a 5-handoff call).
- **Interference:** C/I (co-channel only) = 15 dB (just meets target). Adding adjacent-channel power: I_{total} = −83.81 dBm, SINR = **13.81 dB** → target of 15 dB **not met** (shortfall 1.19 dB). Mitigation: 120° sectoring with down-tilt (≈ +4.8 dB).
- **Spectrum bands (100 m):** FSPL = 71.53 dB (900 MHz), 83.32 dB (3.5 GHz, +11.80 dB), 101.38 dB (28 GHz, +29.86 dB). Path loss alone does not set performance: bandwidth, beamforming gain, density and SINR matter too.
- **Coordination:** ITU-R develops the framework, WRC revises the Radio Regulations, BR records and coordinates assignments; national regulators and operators must follow the resulting allocations and cross-border coordination.
- **Recommendation:** a staged plan – (1) sector and optimise the 900 MHz / LTE macro layer, (2) split cells and add 3.5 GHz NR where demand exceeds ~88 E per 100-channel group, (3) 28 GHz small-cell hot-spots, (4) 6G-ready architecture.

Given Data and Engineering Assumptions

Quantities not stated in the problem are assumed below. All calculations use these values so the results are reproducible. Constants: $c = 3 \times 10^8$ m/s; $FSPL(dB) = 20 \log_{10}(4\pi d/\lambda)$.

Parameter	Value	Source / reason
Cell geometry	Hexagonal; r = 2 km; 19-cell planning cluster; N = 7	Given
Traffic	1000 subscribers × 0.08 E; C = 100 channels; blocking ≤ 2 %	Given
Mobility	60 km/h; overlap dwell 6 s; route cell length 1 km; interruption 0.5 s; target cell unavailable 1 %	Given
Interference	S = −70 dBm; I_1 = −85 dBm (co-channel); I_2 = −90 dBm (adjacent); target SINR 15 dB	Given
Spectrum comparison	900 MHz, 3.5 GHz, 28 GHz at d = 100 m	Given
1G idealization	Guard bands, control channels and FDD duplex split are ignored for the 333-channel figure	Stated as required
Traffic model	Poisson call arrivals, exponential holding times, blocked calls cleared (Erlang B), all 1000 users offered to one pooled channel group	Assumed (standard Erlang B)
Route length	1 km = distance travelled along the route between successive handoffs (not 2r = 4 km)	Assumed interpretation
Noise	Thermal noise floor not given, so SINR = $S/(I_1+I_2)$; a real SINR would be slightly lower	Assumed
Propagation	Free space for the band comparison; path-loss exponent n = 4 for reuse/sectoring estimates	Assumed (urban macro)
Handoff failure	Failure probability = probability the target cell has no free channel (1 %); independent attempts	Assumed

STAGE 1 – Explain the Evolution of Mobile Systems (15 marks)

The requirement is to trace 1G to the 6G vision and relate each step to spectrum, capacity and interference management. The comparison below uses the generational examples in the planning data.

1.1 Generational comparison table

Generation	Access / radio technology	Main service capability	Limitation / reason for change
1G AMPS	Analog FM, FDMA; 800 MHz band; 30 kHz per channel; 10 MHz allocation	Analog voice, basic mobility with handoff	Low capacity (one user per 30 kHz), no encryption, poor quality, large reuse (N = 7) needed for ~18 dB C/I → motivated digital systems

Generation	Access / radio technology	Main service capability	Limitation / reason for change
2G GSM	Digital, TDMA/FDMA; 900 MHz; 200 kHz carrier; 8 slots per carrier	Digital voice, SMS, encryption, international roaming	Circuit-switched, only ~10–14 kbps data → motivated packet data and wider carriers
3G UMTS	WCDMA; 5 MHz carrier; frequency reuse 1; soft handoff	Voice plus mobile data and multimedia	Interference-limited (power control, near-far); data rates and spectral efficiency inadequate for broadband → motivated OFDM/MIMO
4G LTE	OFDMA (downlink) / SC-FDMA (uplink), MIMO; all-IP packet core	Mobile broadband, video, low-latency IP services	Sub-6 GHz spectrum scarce; not designed for massive IoT or ultra-reliable low latency → motivated new bands and service classes
5G NR	OFDM-based NR; eMBB, URLLC, massive IoT; mid-band and mmWave; beamforming	Multi-Gbps broadband, ~1 ms-class latency services, dense IoT	mmWave range and blockage, dense small-cell cost, energy → motivated AI-driven, sensing-aware network
6G (vision)	THz, AI-native networking, ISAC (integrated sensing and communication)	Holographic communication, tactile internet, sensing as a service	Not yet standardized; THz range, hardware, energy and spectrum availability are open issues

1.2 Wavelength and idealized channel count

$\lambda = c / f = (3 \times 10^8 \text{ m/s}) / (800 \times 10^6 \text{ Hz}) =$ ***0.375 m***

$N_{ch} = B_{total} / B_{ch} = 10 \text{ MHz} / 30 \text{ kHz} = 333.3 \approx$ ***333 channels***

Practical qualification: this idealized figure ignores guard bands, control/signalling channels and duplexing. If the 10 MHz had to carry both uplink and downlink (FDD), only about 166 duplex pairs would exist, and with N = 7 reuse each cell could use only about 333/7 ≈ 47 channels. The real number is therefore lower than 333.

1.3 2G digital transmission and TDMA versus 1G analog

A GSM carrier of 200 kHz carries 8 TDMA slots, i.e. 200/8 = 25 kHz per slot, against 30 kHz per analog 1G channel. In 10 MHz the idealized GSM capacity is (10 MHz/200 kHz) × 8 = 50 × 8 = 400 slots versus 333 analog channels, a 1.2× gain before any other benefit. The larger practical gain comes from digital speech coding, error-correcting codes, interleaving and equalization: digital links tolerate a lower C/I (about 9–12 dB instead of about 18 dB), so a smaller reuse cluster can be used, and half-rate coding can double the slot count again. Digital transmission also adds encryption, SMS and better quality.

1.4 Synthesis

Across the generations the scarce resource moved from “channels” to “spectrum efficiency”. In 1G, capacity was bought almost only by adding spectrum and cells, and interference was managed by a fixed reuse pattern (N = 7). 2G made the radio link robust against interference,

so the same spectrum could be reused more tightly. 3G removed frequency planning (reuse 1) but made interference a shared, power-controlled quantity. 4G used OFDMA and MIMO to schedule resources adaptively, and 5G adds wide mid-band and mmWave spectrum, beamforming and network slicing, so interference is now managed in space (beams) as well as in frequency and time. The 6G vision extends this with THz bands, AI-based resource control and sensing-assisted networking. The planning consequence is that every later generation needs more cells, more bandwidth and smarter interference control, which is exactly the trade-off examined in Stages 2–6.

STAGE 2 – Analyze Cell Coverage and Frequency Reuse (15 marks)

2.1 Cell area and cluster area

$A_{cell} = (3\sqrt{3} / 2)\, r^2 = 2.598 \times (2\ km)^2 =$ ***10.39 km²***

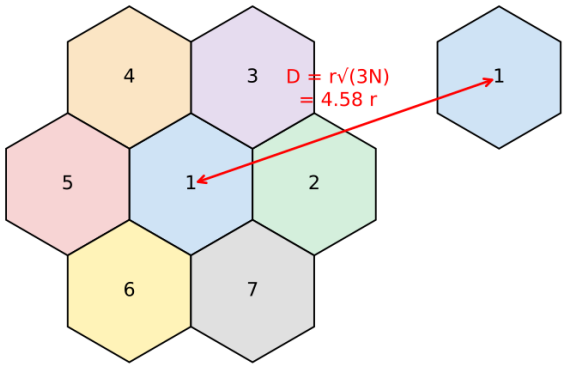
$A_{19} = 19 \times 10.39\ km^2 =$ ***197.45 km²***

2.2 Reuse fraction and reuse distance (N = 7)

Reuse fraction = 1/N = 1/7 = ***0.143*** *(each cell may use about 14.3 % of the channels)*

$D = r\sqrt{(3N)} = 2 \times \sqrt{21} =$ ***9.17 km***; $D/r = \sqrt{(3N)} = 4.58$

Fig. 1 Seven-cell reuse cluster (N = 7) and nearest co-channel cell



Assumptions of the hexagonal model: flat terrain, identical omnidirectional base stations at cell centres, uniform power and traffic, ideal circular propagation boundary approximated by a hexagon (which tiles the plane without gaps or overlaps), and fixed identical cell size. The hexagon is a planning abstraction, not a measured footprint.

Geometric coverage vs guaranteed radio coverage: the hexagon only says which base station is nearest. Guaranteed radio coverage needs a link budget: received power above sensitivity with fading margin, shadowing margin, building penetration loss and an acceptable SINR. Terrain, clutter, antenna height, shadowing and interference make the real footprint irregular, so edge users inside the hexagon may still have poor service.

2.3 Capacity-expansion techniques

Technique	Coverage effect	Capacity effect	Interference / handoff

Technique	Coverage effect	Capacity effect	Interference / handoff
Cell splitting	Cell radius reduced (e.g. 2 km → 1 km), same area needs 4× cells; power reduced ≈ 12 dB for n = 4	Up to 4× capacity per area with same channel set	D/r unchanged so C/I preserved; handoff rate doubles
Sectoring (120°)	Same footprint, directional antennas	Same spectrum, interferers cut from 6 to 2 (≈ +4.8 dB C/I) allowing smaller N	Less co-channel interference; more intra-site handoffs; trunking loss
Microcells	Radius of a few hundred metres, street level	High capacity in hot-spots	Contained interference; very frequent handoff for fast users
Picocells	Indoor / venue, tens of metres	High capacity density	Low leakage; isolated by walls
Femtocells	Home / office, ~10 m, user-installed	Offload from macro; good indoor service	Can interfere with macro if not coordinated; handoff in/out of cell

Recommendation for the regional operator: use 120° sectoring with down-tilt on the existing macro sites first (cheap, no new sites, and it also fixes the interference shortfall found in Stage 5), then selective cell splitting or microcells only in the cells whose busy-hour traffic approaches the ~88 E limit of a 100-channel group (Stage 3). Pico/femto cells then serve indoor hot-spots. **Effects:** sectoring lowers co-channel interference but adds sector-to-sector handoffs and reduces trunking efficiency; splitting keeps C/I but doubles the handoff rate (cell crossing time falls from 60 s to about 30 s at 60 km/h) and raises site and backhaul cost.

STAGE 3 – Estimate Traffic Capacity (15 marks)

3.1 Offered traffic

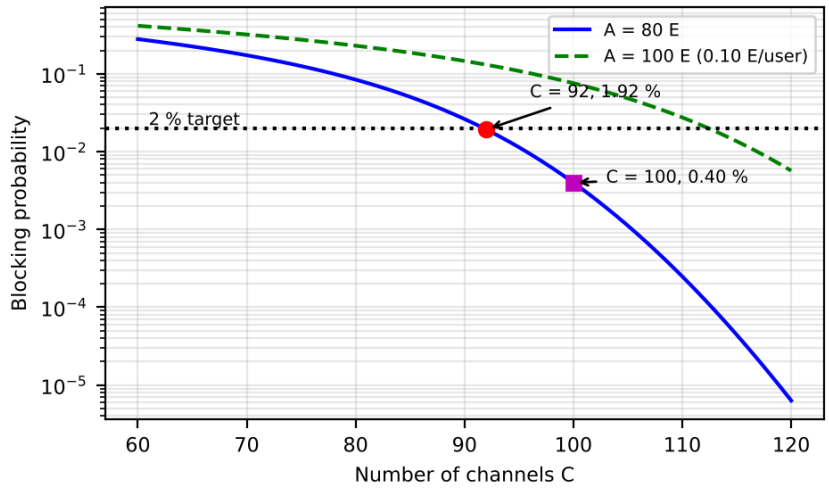
$A = N_{users} \times a = 1000 \times 0.08\ E =$ ***80 Erlang***

3.2 Erlang B model

$B(A, C) = (A^C/C!) / \Sigma_{k=0..C}\ (A^k/k!)$, *computed recursively as* $B(A,k) = A \cdot B(A,k-1) / [k + A \cdot B(A,k-1)]$, $B(A,0) = 1$

Blocking with C = 100: B(80, 100) = **0.399 % ≈ 0.40 %**, well under the 2 % target. **Smallest C meeting 2 %:** B(80, 91) = 2.25 % (fails) and B(80, 92) = 1.92 % (passes), so **C = 92 channels**.

Fig. 2 Erlang B blocking versus channel count



Quantity	Value	Meaning
Offered traffic A	80 E	Average busy-hour channel demand
Blocking, C = 100	0.40 %	Comfortable margin; about 8 spare channels at the 2 % level
Minimum C for ≤ 2 %	92 channels (1.92 %)	C = 91 gives 2.25 %, so 92 is the threshold
Carried traffic at C = 92	80 × (1 − 0.0192) = 78.5 E	Average busy channels; occupancy per channel = 85.3 %
100-channel groups needed	⌈92 / 100⌉ = **1**	Single group covers the pooled demand
Maximum load of 100 channels at 2 %	≈ 88 E (0.088 E/user)	Only about 10 % growth headroom

Why this is only a first-order estimate: (i) it pools all 1000 users into one group, but the 19-cell plan has demand spread over many cells; at an average of 80/19 = 4.2 E per cell, each cell would need about 9 channels for 2 % blocking, i.e. about 171 channels in total instead of 92 (loss of trunking efficiency); (ii) with N = 7 reuse only about one-seventh of the spectrum is available per cell; (iii) control channels, sectors, handoff reservation, uneven (hot-spot) traffic, coverage, interference, backhaul and site count are not modelled. It is therefore a sizing indicator, not a site plan.

Trade-offs: adding channels reduces blocking but with diminishing returns and higher spectrum/equipment cost; accepting a higher blocking target saves infrastructure but loses revenue and user satisfaction; large pooled groups are more efficient than many small ones, so sectoring and splitting improve capacity per area while lowering trunking efficiency. The sensitivity is large: at 0.10 E/user the demand is 100 E and needs 113 channels (two groups).

Assumptions behind the offered-traffic model: Poisson (random) call arrivals, exponentially distributed holding times, blocked calls cleared, identical users with a constant 0.08 E in the busy hour, and no retry traffic.

STAGE 4 – Evaluate Mobility and Handoff (15 marks)

4.1 Distance in the overlap interval

v = 60 km/h = 60 / 3.6 = 16.67 m/s; d = v × t = 16.67 × 6 s = ***100 m***

A handoff decision must be completed within a 100 m overlap zone, i.e. 10 % of the 1 km route cell length. Faster users or smaller cells leave less time for measurement and signalling.

4.2 Handoff rate

Interpretation: the 1 km value is taken as the distance a mobile travels along the route within one cell between successive handoff points (a chord through the cell), not the 4 km cell diameter. Then:

t_{cell} *= 1000 m / 16.67 m/s = 60 s; handoff rate = 60 s / 60 s =* ***1 hard handoff per minute***

If the route passed through the cell centre with the 2 km radius, the length would be up to 4 km and the rate would fall to 0.25 per minute, so the result depends on the interpretation and on the real route geometry.

4.3 Hard, soft and softer handoff

Type	How it works	Connection continuity	Resource use
Hard	Break-before-make: old link released, then new link set up (AMPS, GSM, inter-frequency LTE)	Short interruption (0.5 s here); risk of drop if target has no channel	Lowest: only one channel at a time
Soft	Make-before-break: mobile connected to two base stations simultaneously (CDMA/UMTS)	Seamless; diversity gain	Higher: two channels/backhaul legs during overlap
Softer	Between sectors of the same base station, combined at that site	Seamless	Moderate: extra sector resource but no extra backhaul

4.4 Simple call-drop risk for one handoff

The handoff fails if the target cell has no free channel, with probability p = 0.01. A first-order drop risk is therefore:

$P_{drop} \approx p =$ ***0.01 (1 %)*** *per hard handoff; over a call with 5 handoffs:* $1 - 0.99^5 = 4.9\ \%$

The 0.5 s interruption is only 0.5/6 = 8.3 % of the 6 s overlap time, so in principle several attempts can be made before the link fails, and the interruption is 0.5/60 = 0.83 % of the time spent in a cell. **Limits of the estimate:** it assumes independent attempts with a fixed 1 % unavailability (in reality blocking is correlated in time and heavier in busy cells), and it ignores signal fading and measurement delay, ping-pong handoffs, speed variation, handoff-priority (guard channel) schemes, and the call-completion effect of the interruption itself.

STAGE 5 – Quantify Interference and Compare Spectrum Bands (20 marks)

5.1 Carrier-to-interference ratio (co-channel only)

$C/I = S - I_1 = -70 - (-85) =$ ***15 dB*** *→ just equal to the 15 dB target (zero margin).*

5.2 Total interference and SINR

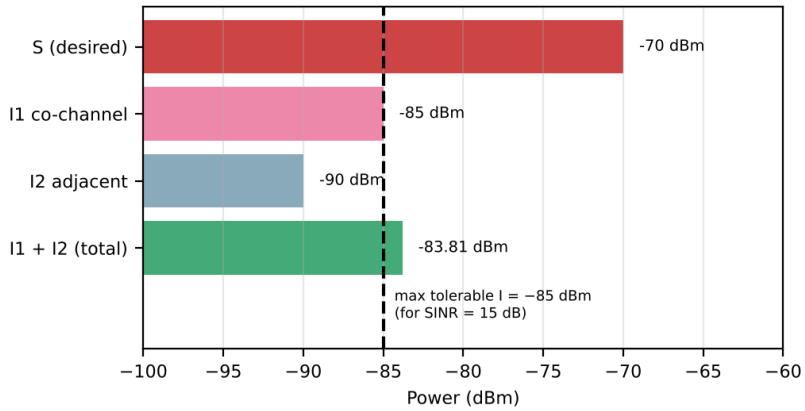
Interference powers must be added in linear units, not in dB:

$I_1 = 10^{-8.5}\ mW = 3.162 \times 10^{-9}\ mW;\quad I_2 = 10^{-9}\ mW$

$I_{total} = 4.162 \times 10^{-9}\ mW = 10\ log_{10}(4.162 \times 10^{-9}) = \mathbf{-83.81\ dBm}$

$SINR = S - I_{total} = -70 - (-83.81) = \mathbf{13.81\ dB}$

Fig. 3 Signal and interference levels

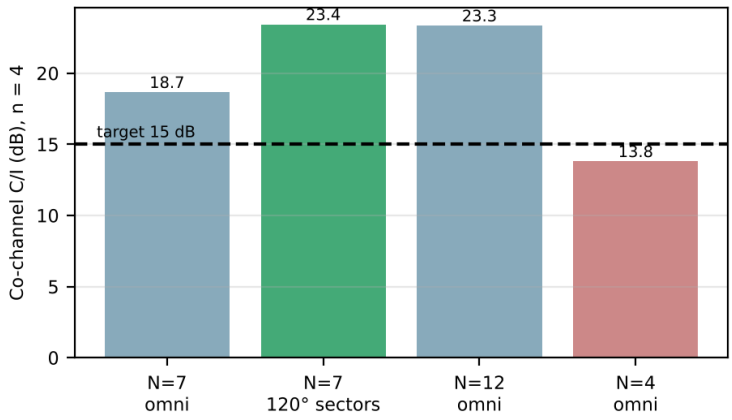


Decision: 13.81 dB < 15 dB, so the target is **not met** (shortfall 1.19 dB). The adjacent-channel signal, although 5 dB weaker than the co-channel one, adds 1.19 dB of interference. Thermal noise was not given; including it would lower the SINR further.

5.3 Mitigation and trade-off

Recommended mitigation: 120° sectoring with antenna down-tilt (and power control). Under an n = 4 model the number of first-tier co-channel interferers falls from 6 to 2, an improvement of 10 log(6/2) = 4.8 dB, which exceeds the 1.19 dB needed and leaves about 3.6 dB margin (SINR ≈ 18.6 dB). Alternatives are a larger reuse size (N = 12 gives D/r = 6 and about 23 dB C/I) which costs 42 % of the channels per cell, or better adjacent-channel filtering (ACLR), which costs hardware.

Fig. 5 Effect of reuse size and sectoring on C/I



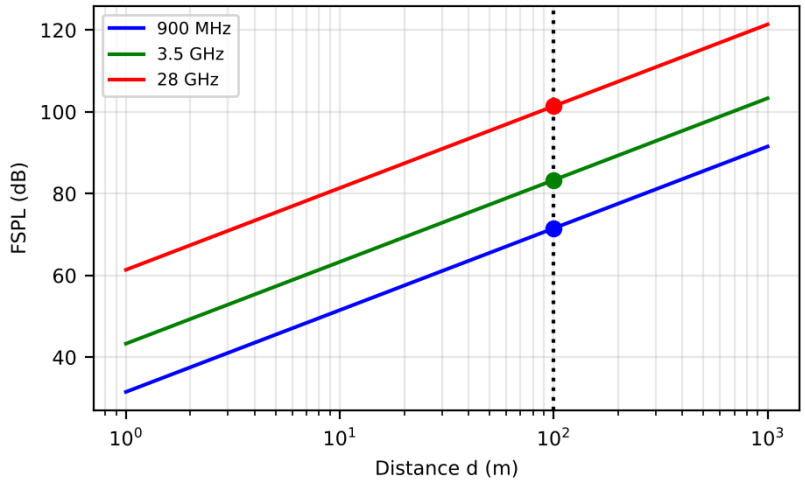
Trade-off: sectoring needs more antennas and radios, produces more (softer/inter-sector) handoffs, and splits each cell's channels into three smaller pools, reducing trunking efficiency (Stage 3).

5.4 Wavelength and free-space path loss at 100 m

$\lambda = c/f;\quad FSPL = 20\ log_{10}(4\pi d/\lambda),\ with\ 4\pi d = 1256.6\ m$

Band	λ	FSPL at 100 m	Δ vs 900 MHz	Half-wave antenna	Penetration	Coverage / capacity role
900 MHz	0.333 m	71.53 dB	0 dB	≈ 16.7 cm	Best (good indoor and foliage)	Wide-area coverage, limited bandwidth (≈ 10–20 MHz)
3.5 GHz	85.7 mm	83.32 dB	+11.80 dB	≈ 4.3 cm	Moderate	Capacity layer, 100 MHz carriers, massive MIMO feasible
28 GHz	10.7 mm	101.38 dB	+29.86 dB	≈ 5.4 mm	Poor (blocked by walls, foliage, bodies)	Very high capacity hot-spots, 400 MHz carriers, many-element arrays

Fig. 4 Free-space path loss versus distance



Why path loss alone does not determine network performance: (i) FSPL grows with f^2 only for a fixed-aperture-gain isotropic antenna; because antenna size shrinks with λ, the same physical area holds many more elements, so beamforming gain (about 10 log of the element count) can recover much of the loss; (ii) the bandwidth available is far larger at higher bands, and capacity follows Shannon's $C = B \log_2(1 + SINR)$: 400 MHz at 10 dB gives about 1.38 Gbit/s, while 20 MHz at 20 dB gives about 133 Mbit/s; (iii) narrow beams and small cells reduce interference and enable spatial reuse; (iv) actual range also depends on transmit power (EIRP), receiver noise, blockage, rain and foliage loss, and cell density; (v) the free-space value is the minimum loss, and real urban propagation is worse for all bands, especially for mmWave.

STAGE 6 – Spectrum Coordination and Deployment Recommendation (20 marks)

6.1 Roles in international spectrum coordination

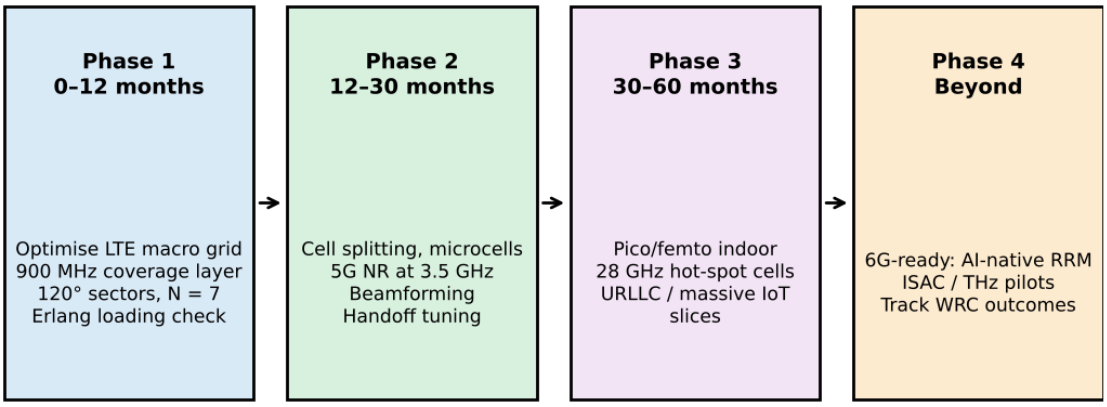
Body	Role

Body	Role
ITU-R (ITU Radiocommunication Sector)	The UN agency's sector responsible for the global use of the radio-frequency spectrum and satellite orbits; produces the Radio Regulations framework, technical Recommendations and reports, and the IMT framework (IMT-2000, IMT-Advanced, IMT-2020, IMT-2030) that defines the generations.
WRC (World Radiocommunication Conference)	Treaty-level conference held every 3–4 years that reviews and revises the Radio Regulations: the table of frequency allocations to services in Regions 1, 2 and 3, identification of bands for IMT, and technical/regulatory conditions. It also sets the agenda for the next conference.
BR (Radiocommunication Bureau)	The executive and administrative arm of ITU-R: processes notifications and coordination requests, maintains the Master International Frequency Register, examines assignments for harmful-interference conformity, publishes information and supports the conferences and the Radio Regulations Board.

How coordination affects national regulators and operators: a national regulator must align its band plan and licences with ITU allocations and, near borders, coordinate with neighbouring administrations so that licensed use does not cause harmful interference across the border. For operators this means that the spectrum they can obtain, its technical limits (power, out-of-band emission) and its cross-border usability are decided largely in advance, while globally harmonized bands (such as the widely used 3.5 GHz range and the mmWave bands identified for IMT) give larger device ecosystems, lower equipment cost and easier roaming. The 28 GHz portion must be checked against the national allocation, because mmWave allocations differ between countries.

6.2 Staged deployment strategy

Fig. 6 Staged deployment strategy



Phase	Action	Evidence from the analysis
1 (0–12 months)	Optimise the existing LTE/macro grid on 900 MHz as the coverage layer: 120° sectoring, down-tilt, power control, N = 7 retained; check Erlang loading	SINR 13.81 dB → ≈ 18.6 dB (needs ≥ 15 dB); 92 channels needed ≤ 100-channel group; handoff 1/min is manageable with hard handoff

Phase	Action	Evidence from the analysis
2 (12–30 months)	Cell splitting and microcells in dense areas; add 5G NR at 3.5 GHz as capacity layer co-sited with macro sites, using beamforming	Hot-spot cells reaching ~88 E; +11.8 dB path loss vs 900 MHz is recoverable with massive-MIMO beam gain; handoff rate doubles to ~2/min and needs tuned thresholds (100 m overlap)
3 (30–60 months)	Pico/femto and 28 GHz mmWave small cells for stadiums, malls, campuses; URLLC/massive-IoT slices	+29.9 dB path loss, poor penetration → LOS small cells with high-gain arrays; 400 MHz-class bandwidth gives Gbit/s capacity
4 (beyond)	6G-ready architecture: AI-based radio resource management, ISAC pilots, monitor WRC outcomes for new bands	Evolution trend (Stage 1): interference/capacity management moves to intelligent, dynamic control

Integration: the evolution from 1G to 5G shows that capacity growth comes from three levers – more spectrum (3.5 GHz, 28 GHz), more spatial reuse (sectors, small cells) and better spectral efficiency (OFDMA/MIMO/beamforming) – and each lever increases interference management and handoff workload. The calculated plan (80 E, 92 channels, N = 7, D = 9.17 km, 1 handoff per minute, SINR 13.81 dB) shows the current macro design has little margin: the interference result fails and traffic headroom is only ~10 %, so improving SINR and splitting come before adding new bands.

Key assumptions and uncertainties to verify before deployment:

- Actual busy-hour traffic per user (0.08 E vs 0.10 E changes the requirement from 92 to 113 channels) and its spatial distribution over the 19 cells.
- Real propagation (path-loss exponent, shadowing, penetration loss) and the true cell radius and route geometry (meaning of the 1 km length).
- Noise floor, adjacent-channel rejection and the actual co-channel interferer count; the 13.81 dB SINR excludes thermal noise.
- Handoff target availability, handoff thresholds, ping-pong behaviour and guard-channel policy.
- National availability, licence conditions and cross-border coordination of 900 MHz, 3.5 GHz and 28 GHz; device ecosystem; backhaul, site-acquisition cost and energy budget.

CALCULATION VERIFICATION – Python Script and Results

All numerical results were checked with the short script below (the same equations as the hand calculations; Erlang B uses the standard recursion). The output matches the hand values.

```
# CO1 - Wireless planning calculations (Python 3)
import math
c = 3e8
# Stage 1
lam800 = c/800e6;  n_ch = 10e6/30e3
print(f"lambda(800 MHz) = {lam800:.3f} m ; channels = {n_ch:.1f}")
print(f"GSM slots in 10 MHz = {10e6/200e3*8:.0f}")
# Stage 2
r, N = 2.0, 7
A_cell = 1.5*math.sqrt(3)*r**2
print(f"Cell area = {A_cell:.2f} km2 ; 19 cells = {19*A_cell:.2f} km2")
print(f"Reuse fraction = 1/{N} = {1/N:.4f}")
print(f"D = {r*math.sqrt(3*N):.2f} km ; D/r = {math.sqrt(3*N):.2f}")
# Stage 3  (Erlang B recursion)
def erlang_b(A, C):
    b = 1.0
    for k in range(1, C+1):
        b = A*b/(k + A*b)
    return b
A = 1000*0.08
print(f"Offered traffic A = {A:.0f} E ; B(80,100) = {erlang_b(A,100)*100:.3f} %")
C = 1
while erlang_b(A, C) > 0.02: C += 1
print(f"Smallest C for B<=2% : {C} (B = {erlang_b(A,C)*100:.2f} %)")
print(f"B(C-1) = {erlang_b(A,C-1)*100:.2f} %")
print(f"Groups of 100 channels = {math.ceil(C/100)} ; carried = {A*(1-erlang_b(A,C)):.1f}
E")
# Stage 4
v = 60/3.6
print(f"v = {v:.2f} m/s ; distance in 6 s = {v*6:.0f} m")
print(f"time per 1 km = {1000/v:.0f} s ; handoffs/min = {60/(1000/v):.2f}")
p = 0.01
print(f"P_drop(1 handoff) = {p:.3f} ; 5 handoffs = {1-(1-p)**5:.4f}")
print(f"interruption/overlap = {0.5/6:.3f}")
# Stage 5
S, I1, I2 = -70, -85, -90
mw = lambda x: 10**(x/10)
I_tot = 10*math.log10(mw(I1)+mw(I2))
print(f"C/I = {S-I1} dB ; I1+I2 = {mw(I1)+mw(I2):.3e} mW = {I_tot:.2f} dBm")
print(f"SINR = {S-I_tot:.2f} dB")
f0 = None
```

```
for f in (900e6, 3.5e9, 28e9):
    lam = c/f; L = 20*math.log10(4*math.pi*100/lam)
    f0 = f0 or L
    print(f"f = {f/1e9:5.2f} GHz  lambda = {lam*1e3:7.2f} mm  "
          f"FSPL(100 m) = {L:6.2f} dB  delta = {L-f0:5.2f} dB")
```

Script output:

```
lambda(800 MHz) = 0.375 m ; channels = 333.3
GSM slots in 10 MHz = 400
Cell area = 10.39 km2 ; 19 cells = 197.45 km2
Reuse fraction = 1/7 = 0.1429
D = 9.17 km ; D/r = 4.58
Offered traffic A = 80 E ; B(80,100) = 0.399 %
Smallest C for B<=2% : 92 (B = 1.92 %)
B(C-1) = 2.25 %
Groups of 100 channels = 1 ; carried = 78.5 E
v = 16.67 m/s ; distance in 6 s = 100 m
time per 1 km = 60 s ; handoffs/min = 1.00
P_drop(1 handoff) = 0.010 ; 5 handoffs = 0.0490
interruption/overlap = 0.083
C/I = 15 dB ; I1+I2 = 4.162e-09 mW = -83.81 dBm
SINR = 13.81 dB
f =  0.90 GHz  lambda =  333.33 mm  FSPL(100 m) =  71.53 dB  delta =  0.00 dB
f =  3.50 GHz  lambda =   85.71 mm  FSPL(100 m) =  83.32 dB  delta = 11.80 dB
f = 28.00 GHz  lambda =   10.71 mm  FSPL(100 m) = 101.38 dB  delta = 29.86 dB
```

Comparison of hand calculation with script: all quantities agree exactly because the same closed-form equations are used; the script adds the Erlang B recursion and the threshold search, which are impractical by hand. The interpretation of each result is given in Stages 1–6.

FINAL SYSTEM VERIFICATION

Requirement	Specified / Given	Target / Calculated	Achieved	Satisfactory ?
Wavelength at 800 MHz	800 MHz	0.375 m	0.375 m	Yes
Idealized 1G channels	10 MHz / 30 kHz	333	333 (guard bands, duplex ignored)	Yes (idealized)
Cell area / 19-cell area	r = 2 km	10.39 / 197.45 km²	10.39 / 197.45 km²	Yes
Reuse (N = 7)	N = 7	1/7; D = r√21	0.143; 9.17 km	Yes
Offered traffic	1000 × 0.08 E	80 E	80 E	Yes
Blocking, C = 100	≤ 2 %	Erlang B	0.40 %	Yes
Minimum channels for 2 %	≤ 2 %	92	92 (1.92 %); 1 group of 100	Yes
Overlap distance	60 km/h, 6 s	100 m	100 m	Yes
Handoff rate	1 km cell length	1 per min	1 per min	Yes (interpretation stated)
Call-drop risk	1 % unavailable	≈ 1 % per handoff	1 % (4.9 % over 5)	Acceptable, with caveats
Co-channel C/I	≥ 15 dB	15 dB	15 dB	Marginal (zero margin)
SINR with adjacent channel	≥ 15 dB	13.81 dB	13.81 dB; ≈ 18.6 dB after 120° sectoring	**No → Yes after mitigation**
FSPL at 100 m	900 MHz / 3.5 GHz / 28 GHz	71.53 / 83.32 / 101.38 dB	Δ = +11.80 / +29.86 dB	Yes
Spectrum coordination roles	ITU-R, WRC, BR	Described	Described with national impact	Yes
Integrated recommendation	Staged plan	Phases 1–4	Linked to all calculations	Yes (conditional on assumptions)

The one requirement that fails in the supplied scenario is the 15 dB SINR target (13.81 dB), and the recommended 120° sectoring with down-tilt closes this gap with margin. The traffic requirement is met with only about 10 % headroom, which is the main driver for the capacity-expansion phase.

FINAL SYSTEM DESIGN – Integrated Regional Network Plan

Item	Adopted design / value
Cell layout	Hexagonal macro cells, r = 2 km (10.39 km² each), 19-cell planning area (197.45 km²); N = 7, D = 9.17 km

Item	Adopted design / value
Antenna / interference	120° sectors with down-tilt and power control; target SINR ≥ 15 dB (≈ 18.6 dB expected)
Traffic / channels	80 E offered; 92 channels needed for 2 % blocking; one 100-channel group (0.40 % blocking); trigger for expansion at ~88 E
Handoff	Hard handoff at 1 per minute for 60 km/h users; 100 m overlap zone; guard channels for handoff calls; softer handoff between sectors
Capacity expansion	Selective cell splitting and microcells; pico/femto for indoor; 3.5 GHz NR capacity layer; 28 GHz small-cell hot-spots
Spectrum	900 MHz coverage; 3.5 GHz capacity; 28 GHz hot-spot; licences aligned with ITU/WRC allocations and cross-border coordination
Evolution	LTE → 5G NR (eMBB/URLLC/mMTC) → 6G-ready (AI-native, ISAC)

FINAL ENGINEERING DECISION

- **Evolution and spectrum:** adopt a layered 4G/5G network rather than a new single-technology build; each generation adds capacity by spectrum, spatial reuse or efficiency, and the operator should use all three in phases.
- **Cell and traffic plan:** keep N = 7 and r = 2 km macro cells initially; 80 E and 92 channels fit within one 100-channel group but with only ~10 % headroom, so plan splitting/microcells for hot-spots.
- **Handoff:** 1 handoff per minute with 100 m overlap is manageable; handoff rate doubles after splitting, so reserve guard channels and tune thresholds.
- **Interference:** the supplied case fails the 15 dB target (13.81 dB); implement 120° sectoring with down-tilt first, accepting the cost of more handoffs and trunking loss.
- **Spectrum bands:** 900 MHz for coverage, 3.5 GHz for capacity with massive MIMO, 28 GHz for hot-spots; path loss differences (+11.8 dB, +29.9 dB) are offset by beamforming and bandwidth rather than ignored.
- **Coordination and risks:** confirm national allocations, cross-border coordination, real traffic, propagation and noise measurements, and the cost/energy of densification before commitment.

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